INCH-POUND

A-A-50622
14 March 1994

COMMERCIAL ITEM DESCRIPTION

SURGE SUPPRESSOR FOR SHIPBOARD PERSONAL COMPUTERS AND PERIPHERAL EQUIPMENTS

The General Services Administration has authorized the use of this Commercial Item Description.

Abstract. This Commercial Item Description covers cord-connected surge suppressors with or without individual switches for receptacles. These transient voltage surge suppressors (TVSS) are intended to limit the maximum amplitude of transient voltage surges on shipboard power lines that supply personal computers and other peripheral equipment.

These CID requirements cover hybrid surge suppressors: an assembly of series and parallel, or any combination of suppression elements with compatible filter devices to limit voltages, current or both to a specified level. The hybrid TVSS shall include sine wave clamping capability, multiple stage surge clamping, and an EMI/RFI noise filtering network.

These transient voltage surge suppressors are intended to be installed on the load side of the ship's branch overcurrent protection device and operate on an ungrounded delta electrical system. They are not intended to function as surge arrestors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6150

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Salient characteristics. The product shall be certified by a nationally recognized testing laboratory (NRTL) accredited for TVSS certification by the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor for compliance with UL Standard of Safety "Transient Voltage Surge Suppressors", UL 1449 (including any Marine supplements) and the requirements specified herein.

1. The TVSS shall be designed to limit the transient voltage surge to a maximum amplitude when subjected to a 1.2 by 50.0 microsecond 6000 volt, 500 ampere ringwave surge.

2. Types. The surge suppressor shall be a two-port device per UL 1449 and shall be one of the following types.

   a. Cord-connected

   b. Power control-center (Power control-center is a TVSS with individual power (on-off) switches for all receptacles.)

3. Protection modes. All surge suppressors are for use in an ungrounded electrical system and shall provide protection for common mode (line-to-ground) and normal mode (line-to-line) transients.

4. Electrical rating. The surge suppressor shall be rated for continuous duty at nominally 115 volts, single phase, 15 amperes.

5. Voltage protection level. Either 330 or 400 volts per UL STD 1449 ratings (all modes).

6. Response time. Shall be less than 50 nanoseconds.

7. Supplementary protection. All surge suppressors shall be furnished with the following:

   a. Circuit breaker. A double-pole circuit breaker shall provide overcurrent protection on both sides of the AC supply line. The breaker shall include a press-to-reset feature.

   b. Temperature protective device. Surge suppressors using MOVs for normal mode protection shall include a non-resettable thermal cutoff device that senses the internal temperature of the enclosure and interrupts both sides of the supply source. The thermal cutoff device shall be rated between 90°C and 100°C.

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c. Surge protection indicator. Surge suppressors shall include a visible indicator to show that the surge suppression circuitry is not damaged.

8. Enclosure. The surge suppressor enclosure shall be metallic.

9. Switches. The following double-pole switches shall be furnished.

a. A main power switch (On-Off) shall be included that de-energizes both lines and the entire surge suppressor unit. The main switch shall be clearly labeled as such. The circuit breaker may be used for this function.

b. For power control-center type TVSS, individual switches (On-Off) shall be provided for each receptacle.

10. Receptacles. Receptacles used in TVSS shall be general purpose, 2 pole, 3 wire, 15 ampere, 125 volts, single or parallel-duplex type. TVSS shall have a minimum of 5 receptacles unless otherwise specified.

11. Cord. The supply cord shall be permanently attached to the suppressor.

a. Flexible cords acceptable are types S, SO, SJ, and SJO.

b. Length of supply cord shall be 6-feet unless otherwise specified in ordering data.

12. Operating/Storage temperature. The TVSS shall be able to operate within a temperature range of -10°C to 60°C. The TVSS must be capable of being stored in a temperature range of -30°C to 70°C without affecting its subsequent operation.

13. Mounting Provisions. Surge suppressors shall include mounting provisions to permit units to be bulkhead, deck or table mounted.

14. Bid samples. When specified in the solicitation, bid samples shall be furnished.

Quality assurance provisions.

1. Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers own drawings,

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specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

2. Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376 and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

3. Regulatory Requirements. The offeror/contractor is encouraged to use recoverable materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Marking on the TVSS shall include "Suitable For Shipboard Use".

Notes. Purchaser should specify type, cord-length, number of receptacles and if bid samples are required.

1. Ordering data. Purchaser shall specify the following:

(a) Title, number and date of this commercial item description.

(b) CID type (see salient characteristics).

(c) Number of receptacles (for cord connected TVSS units; suggested numbers of receptacles are 5, 6 or 7).

(d) Cord length (if other than 6-feet; other suggested lengths are 9-, 12-, and 15-feet).

FED-STD-376 is available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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MILITARY INTERESTS:

Military coordinating activity:
Navy - SH

Preparing activity:
Navy - SH
(Project 6150-N239)

Review activity:
DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-50622	2. DOCUMENT DATE (YYMMDD) 940314

3. DOCUMENT TITLE
SURGE SUPPRESSOR FOR SHIPBOARD PERSONAL COMPUTERS AND PERIPHERAL EQUIPMENTS

4. NATURE OF CHANGE (identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) MR. BRUCE JACKSON, SEA 03E23 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-7440 8-332-7440
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03 42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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